THEORY OF OPERATION

WFCO intelligent product development can be found in a three-stage charging system – a standard feature on all WFCO products. WFCO's automatic three-stage converter handles every charging need for the RV while extending the battery's life by preventing over-charging that can damage the battery.

WFCO Converters of every style have become the favored brand for power conversion and electric distribution in the RV industry. They provide RV owners with an efficient and cost-effective method to use an ac power source and provide power to dc components inside the RV while charging accessory batteries simultaneously.

Basic Operation

RVs are frequently sold with at least one 12 V dc accessory battery installed. This battery is usually a deep-cycle battery that could sustain a slower power drain. RV owners find this practical when loads such as lights, radios, and refrigerators without being connected to ac power or running the motorhome engine. As soon as the RV is connected to ac power, the converter begins changing the battery as needed while at the same time providing 12 V DS power to loads such as lights, radios, and refrigerators.

When the RV is connected to ac power, users frequently use the lights, refrigerators, fans, and other electronics as they would in their homes. RV users also expect the battery to be fully charged when they want to disconnect from power and move the RV or when they are dry camping and turn off their generator.

Tech Support

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WFCO converters are designed to fulfill these needs and expectations by providing three charging stages: absorption, bulk, and float modes.

Absorption mode is the default or regular operation, providing an output of 13.6 volts d[illegible] Because RVs today are designed with converters sized to offer ample dc output power for all dc loads in normal usage, an RV will rarely require anything other than absorption mode.

When a WFCO converter is connected to a battery in absorption mode, power is available for charging that battery whenever the converter output is greater than the voltage level of the battery. If the battery is at or near fully charged, the current drawn from the converter to the battery may be minimal. On the other hand, if the battery were to be fully discharged, the current drawn from the converter to the battery may be high.

Testing has shown that a completely discharged battery of 11.9 V dc connected to a WFCO converter in absorption mode with an output of 13.6 V dc and having a 20 amp lighting load connected to the converter will charge the battery to its fully charged level of 12.7 V dc in fewer than

three hours. Adding more dc loads will lessen the current amount and lengthen the time required to charge the battery. Batteries with damaged cells will also require additional time to charge and may never reach a full charge voltage.

Because of the relationship between voltage and amperage, once the converter reaches its maximumly rated operating current level, any increase in the dc load will start to decrease the voltage output level. When the output from the converter reaches a preset level, the converter will go into bulk mode.

Bulk mode is designed to charge a significantly discharged battery in less time than absorption mode. The microprocessor in WFCO converters continuously monitors the DC line voltage. When the microprocessor detects the preset voltage level, it will boost the converter voltage to 14.4 V dc. The increased voltage will help the battery charge faster while still providing power to the DC appliances in the RV.

In bulk mode, observing the 14.4 V dc output may not be possible because of the voltage-current relationship. To measure the 14.4 V dc, reduce some dc loads while monitoring the voltage at the converter output. As the dc loads are removed, the voltage will begin to climb until 14.4 V dc (nominal) is shown on the meter.

As the battery continues to charge, the current drawn by the battery will gradually decrease. WFCO converters are designed to drop out of bulk mode when the total amperage from the converter reaches a preset point, indicating that the battery is charged. If the amperage draw stays above the preset threshold, the converter will remain in bulk mode for four hours. These features have been implemented to protect and extend the battery's life.

Float mode is the third stage of converter operation. This mode is designed to provide a trickle charge to the battery. If the converter observes no significant variations in the current draw for approximately 44 continuous hours, it will drop the. The converter's output is from 13.6 V to 13.2 V. This lower voltage will keep the battery charged while the RV is not in use. This also helps preserve the battery's life while keeping it charged and ready to use. A change in dc current will cause the converter to exit float mode and return to the default, normal, or absorption mode.

Safety Features
WFCO converters are designed to keep the RV safe and, in some cases, prevent irreparable damage to the converter.

To learn more about our products through video, click here (https://www.youtube.com/channel/UC6ZnA0xjq1EwJRFFeJaer0g/videos). For any WFCO resources, tools, and information on our products, email support@wfcoelectronics.com with your request, or click here (https://wfcoelectronics.com/wp-content/uploads/2022/10/2022-Product-CatalogH.pdf) to download the WFCO Training Catalog.

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